

Deep Learning Lab 1 Report

Fake Review Detection and Product Category Classification

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1 Objective

The objective of this lab is to explore Natural Language Processing techniques using deep learning models. The tasks performed include:

- Text preprocessing using spaCy
- Feature extraction using TF-IDF
- Product category classification using a neural network
- Fake review detection
- Visualization of fake review text using a word cloud

The implementation is performed using Python, PyTorch, and standard NLP libraries.

2 Dataset

The dataset used in this lab consists of product reviews. Each review contains the following information:

- Product category
- Label indicating whether the review is genuine or fake
- Review text

Example product categories include:

- Books
- Electronics
- Clothing
- Home and Kitchen
- Movies and TV

The dataset is preprocessed to remove unnecessary columns and retain only the relevant information required for training the models.

3 Text Preprocessing

Text preprocessing is performed using the spaCy NLP library.

The preprocessing steps include:

- Converting text to lowercase
- Removing stopwords
- Removing punctuation
- Lemmatization
- Keeping only alphabetic tokens

This step reduces noise in the data and improves feature quality for machine learning models.

4 Feature Extraction

After preprocessing, the text data is converted into numerical features using the TF-IDF (Term Frequency–Inverse Document Frequency) representation.

The TF-IDF vectorizer is configured with the following parameters:

- Maximum vocabulary size: 5000
- N-gram range: (1,2)

This means that both unigrams and bigrams are used as features.

5 Task 1: Product Category Classification

In this task, a neural network model is trained to classify product reviews into their respective product categories.

5.1 Model Architecture

The neural network consists of:

- Input layer corresponding to TF-IDF feature dimension
- Fully connected layer with 256 neurons
- Fully connected layer with 128 neurons
- Output layer corresponding to the number of product categories

The model is trained using the Cross Entropy Loss function and the Adam optimizer.

5.2 Training

The model is trained for 10 epochs using mini-batch gradient descent.

The training process gradually reduces the loss and improves classification performance.

5.3 Evaluation Results

The final model achieved an overall accuracy of:

$$Accuracy = 0.7497$$

5.4 Classification Performance

Category	Precision	Recall	F1 Score	Support
Books	0.74	0.73	0.74	874
Clothing Shoes Jewelry	0.78	0.77	0.78	770
Electronics	0.74	0.80	0.77	798
Home and Kitchen	0.70	0.66	0.68	811
Kindle Store	0.79	0.76	0.77	946
Movies and TV	0.90	0.87	0.88	717
Pet Supplies	0.90	0.81	0.85	851
Sports and Outdoors	0.55	0.65	0.59	789
Tools and Home Improvement	0.67	0.68	0.68	772
Toys and Games	0.80	0.78	0.79	759

The macro and weighted averages were approximately:

- Macro Average F1 Score: 0.75
- Weighted Average F1 Score: 0.75

6 Task 2: Fake Review Detection

The next task involves detecting whether a review is genuine or fake.

The label column in the dataset contains:

- CG: Computer Generated (Fake Review)
- OR: Original (Real Review)

The labels are converted into binary values:

- CG $\rightarrow$ 0
- OR $\rightarrow$ 1

A neural network classifier is trained separately for each product category.

6.1 Model Architecture

The fake review classifier consists of:

- Fully connected layer (128 neurons)
- Fully connected layer (64 neurons)
- Output layer for binary classification

The model is trained for 10 epochs with batch size 32.

7 Task 3: Word Cloud Visualization

To understand the common words used in fake reviews, a word cloud is generated using the text of all fake reviews.

The WordCloud library is used to visualize the most frequent words appearing in fake reviews.

The word cloud highlights common patterns in fake reviews such as:

- overly positive language
- repeated promotional phrases
- generic product descriptions

8 Conclusion

In this lab, several NLP and deep learning techniques were applied to analyze product reviews.

Key observations include:

- TF-IDF is an effective method for converting text data into numerical features.
- Neural networks can successfully classify product categories with reasonable accuracy.
- Fake review detection can be formulated as a binary classification problem.
- Visualization techniques such as word clouds help identify common linguistic patterns in fake reviews.

This lab demonstrates how deep learning models can be applied to real-world NLP tasks such as sentiment analysis and fake review detection.